

CoRoT-1b

R-2532

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_200224 · created 2026-08-25T23:16:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458903.6428 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.041
Transit depth (Rp/Rs)^2	2.01 +/- 1.17 [%]
Orbital Inclination (inc)	84.16 +/- 2.83
Airmass coefficient 1 (a1)	0.126 +/- 0.068
Airmass coefficient 2 (a2)	1.49 +/- 1.65
Scatter in the residuals of the lightcurve fit is	47.2 %
Best Comparison Star	None
Optimal Aperture	2.13
Optimal Annulus	7.61
Transit Duration (day)	0.084 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.041 vs 0.1388 (deviation 0.04σ)
Duration fit vs published	0.084 d vs 0.1046 d (deviation 1.1σ)
Mid-time O–C	-10.9 min
Depth SNR	1.9
Residual scatter	47.2 %

Light curve

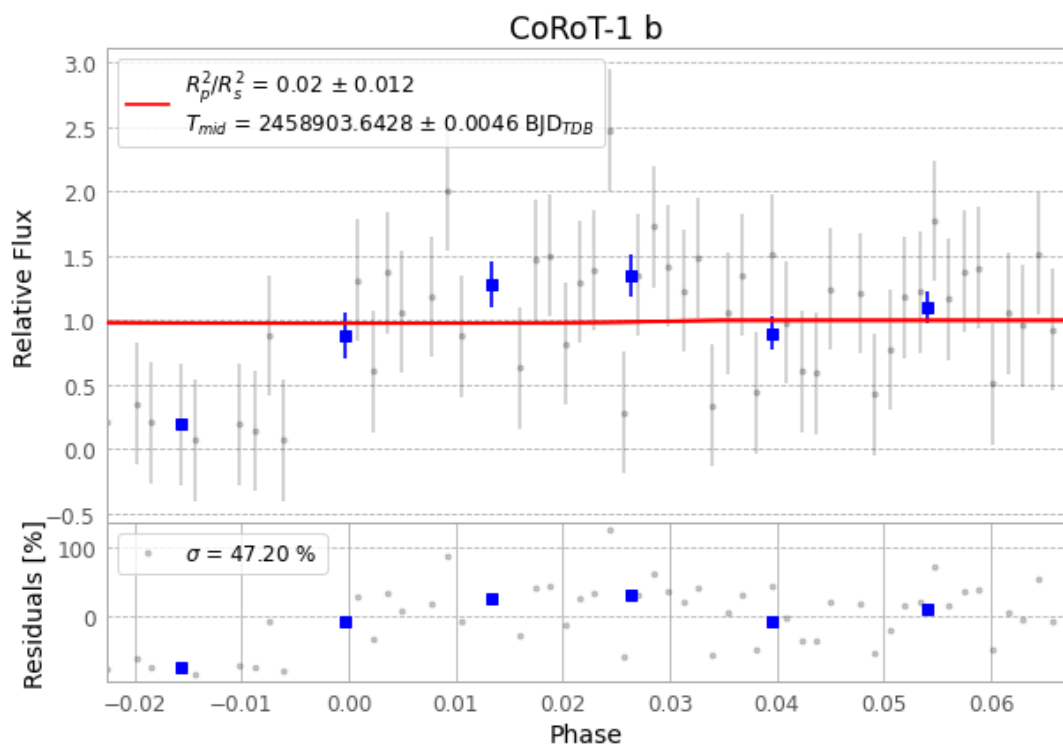


Figure 1 — FinalLightCurve_CoRoT-1 b_2020-02-23.png (SHA-256 52807fe079b8be47...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [174, 431], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcb1c4ae5338ae70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), CoRoT-1200224023012.FITS → CoRoT-1200224054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-200224.log (03138c49be2e...), FinalLightCurve_CoRoT-1 b_2020-02-23.png (52807fe079b8...), FinalLightCurve_CoRoT-1 b_2020-02-23.csv (ea5431775171...)

Compute resources

Wall time 220.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:16Z.